

BCA MATHEMATICS-I (BCA-0101)

Complete Unit-III Study Material

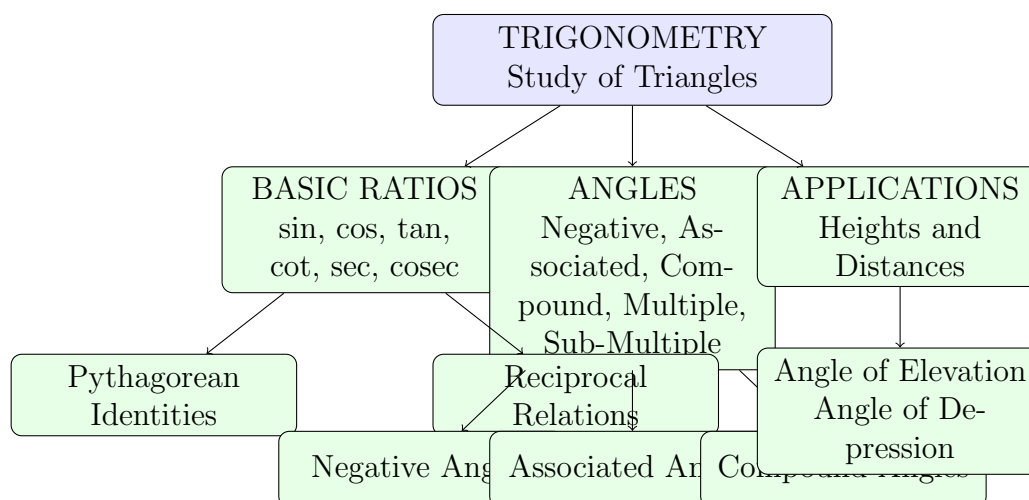
As per Himachal Pradesh University (HPU) BCA Semester-I Syllabus

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Trigonometry Flowchart

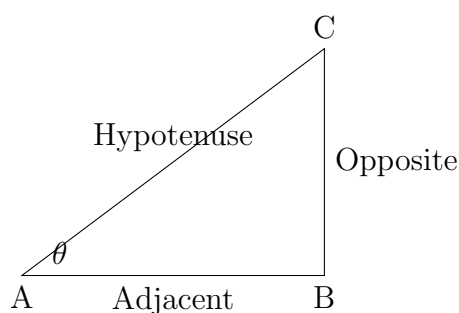


1 Trigonometric Functions

1.1 Definition

Trigonometry deals with the relationships between the angles and sides of triangles.

1.2 Right-Angled Triangle



ABC of the right triangle ($\angle B = 90^\circ$):

Function	Abbreviation	Formula
Sine	$\sin \theta$	Opposite/Hypotenuse = BC/AC
Cosine	$\cos \theta$	Adjacent/Hypotenuse = AB/AC
Tangent	$\tan \theta$	Opposite/Adjacent = BC/AB
Cosecant	$\operatorname{cosec} \theta$	Hypotenuse/Opposite = AC/BC
Secant	$\sec \theta$	Hypotenuse/Adjacent = AC/AB
Cotangent	$\cot \theta$	Adjacent/Opposite = AB/BC

1.3 Reciprocal Relations

$$\begin{aligned}\sin \theta &= \frac{1}{\csc \theta}, & \csc \theta &= \frac{1}{\sin \theta} \\ \cos \theta &= \frac{1}{\sec \theta}, & \sec \theta &= \frac{1}{\cos \theta} \\ \tan \theta &= \frac{1}{\cot \theta}, & \cot \theta &= \frac{1}{\tan \theta}\end{aligned}$$

1.4 Quotient Relations

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

1.5 Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

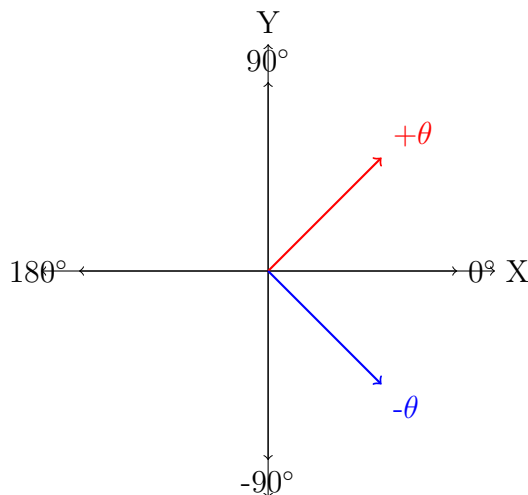
1.6 Standard Angles Table

Angle (θ)	0°	30°	45°	60°	90°
$\sin \theta$	0	1/2	1/√2	√3/2	1
$\cos \theta$	1	√3/2	1/√2	1/2	0
$\tan \theta$	0	1/√3	1	√3	∞
$\cot \theta$	∞	√3	1	1/√3	0
$\sec \theta$	1	2/√3	√2	2	∞
$\operatorname{cosec} \theta$	∞	2	√2	2/√3	1

2 Trigonometric Ratios of Negative Angles

2.1 Concept of Negative Angles

When an angle is measured in **clockwise direction**, it is negative.



2.2 Formulas for Negative Angles

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

$$\cot(-\theta) = -\cot \theta$$

$$\sec(-\theta) = \sec \theta$$

$$(-\theta) = -\theta$$

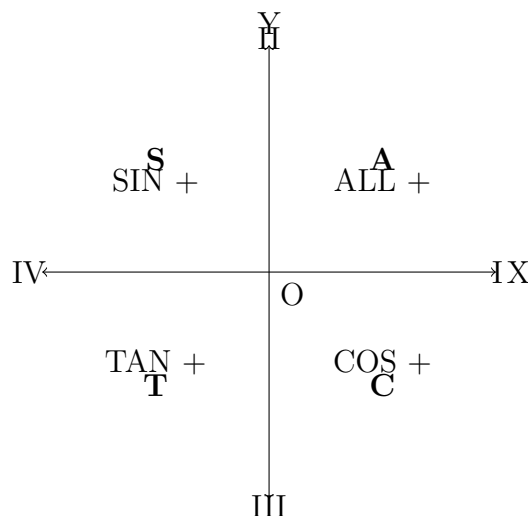
2.3 Important Observations

Function	Effect	Reason
$\sin(-\theta)$	Becomes negative	Sine is odd function
$\cos(-\theta)$	Remains positive	Cosine is even function
$\tan(-\theta)$	Becomes negative	Tangent is odd function

Memory Trick: Only **cos** and **sec** remain positive for negative angles.

3 Trigonometric Ratios of Associated Angles

3.1 Quadrant Rules (ASTC Rule)



ASTC Rule:

- All ratios are positive in **Quadrant I**
- Sine and its reciprocal are positive in **Quadrant II**
- Tangent and its reciprocal are positive in **Quadrant III**
- Cosine and its reciprocal are positive in **Quadrant IV**

3.2 Formulas for Associated Angles

3.2.1 For $(90^\circ - \theta)$ - Complementary Angles:

$$\sin(90^\circ - \theta) = \cos \theta$$

$$\cos(90^\circ - \theta) = \sin \theta$$

$$\tan(90^\circ - \theta) = \cot \theta$$

$$\cot(90^\circ - \theta) = \tan \theta$$

$$\sec(90^\circ - \theta) = \csc \theta$$

$$\csc(90^\circ - \theta) = \sec \theta$$

3.2.2 For $(90^\circ + \theta)$:

$$\sin(90^\circ + \theta) = \cos \theta$$

$$\cos(90^\circ + \theta) = -\sin \theta$$

$$\tan(90^\circ + \theta) = -\cot \theta$$

3.2.3 For $(180^\circ - \theta)$:

$$\sin(180^\circ - \theta) = \sin \theta$$

$$\cos(180^\circ - \theta) = -\cos \theta$$

$$\tan(180^\circ - \theta) = -\tan \theta$$

3.2.4 For $(180^\circ + \theta)$:

$$\sin(180^\circ + \theta) = -\sin \theta$$

$$\cos(180^\circ + \theta) = -\cos \theta$$

$$\tan(180^\circ + \theta) = \tan \theta$$

3.2.5 For $(270^\circ - \theta)$:

$$\sin(270^\circ - \theta) = -\cos \theta$$

$$\cos(270^\circ - \theta) = -\sin \theta$$

$$\tan(270^\circ - \theta) = \cot \theta$$

3.2.6 For $(270^\circ + \theta)$:

$$\sin(270^\circ + \theta) = -\cos \theta$$

$$\cos(270^\circ + \theta) = \sin \theta$$

$$\tan(270^\circ + \theta) = -\cot \theta$$

3.2.7 For $(360^\circ - \theta)$:

$$\sin(360^\circ - \theta) = -\sin \theta$$

$$\cos(360^\circ - \theta) = \cos \theta$$

$$\tan(360^\circ - \theta) = -\tan \theta$$

4 Trigonometric Ratios of Compound Angles

4.1 Definition

Compound Angle: Sum or difference of two or more angles ($A \pm B$).

4.2 Formulas for Compound Angles

4.2.1 Sine Functions:

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

4.2.2 Cosine Functions:

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

4.2.3 Tangent Functions:

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

4.2.4 Cotangent Functions:

$$\cot(A + B) = \frac{\cot A \cot B - 1}{\cot A + \cot B}$$

$$\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$$

4.3 Important Results

4.3.1 Transformations:

$$\begin{aligned} \sin(A + B) \sin(A - B) &= \sin^2 A - \sin^2 B \\ &= \cos^2 B - \cos^2 A \end{aligned}$$

$$\begin{aligned} \cos(A + B) \cos(A - B) &= \cos^2 A - \sin^2 B \\ &= \cos^2 B - \sin^2 A \end{aligned}$$

4.3.2 Special Cases:

$$\begin{aligned} \sin(A + B) + \sin(A - B) &= 2 \sin A \cos B \\ \sin(A + B) - \sin(A - B) &= 2 \cos A \sin B \\ \cos(A + B) + \cos(A - B) &= 2 \cos A \cos B \\ \cos(A - B) - \cos(A + B) &= 2 \sin A \sin B \end{aligned}$$

4.4 Important Values

$$\sin 15^\circ = \frac{\sqrt{3} - 1}{2\sqrt{2}} = \frac{\sqrt{6} - \sqrt{2}}{4}$$

$$\cos 15^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}} = \frac{\sqrt{6} + \sqrt{2}}{4}$$

$$\tan 15^\circ = 2 - \sqrt{3}$$

$$\sin 75^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}} = \frac{\sqrt{6} + \sqrt{2}}{4}$$

$$\cos 75^\circ = \frac{\sqrt{3} - 1}{2\sqrt{2}} = \frac{\sqrt{6} - \sqrt{2}}{4}$$

$$\tan 75^\circ = 2 + \sqrt{3}$$

5 Trigonometric Ratios of Multiple Angles

5.1 Definition

Multiple Angle: $2\theta, 3\theta, 4\theta, \dots (n\theta)$

5.2 Formulas for 2θ (Double Angle)

5.2.1 $\sin 2\theta$:

$$\boxed{\sin 2\theta = 2 \sin \theta \cos \theta}$$

$$\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

5.2.2 $\cos 2\theta$:

$$\boxed{\cos 2\theta = \cos^2 \theta - \sin^2 \theta}$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

$$\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

5.2.3 $\tan 2\theta$:

$$\boxed{\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}}$$

5.2.4 $\cot 2\theta$:

$$\cot 2\theta = \frac{\cot^2 \theta - 1}{2 \cot \theta}$$

5.3 Formulas for 3θ (Triple Angle)

5.3.1 $\sin 3\theta$:

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

5.3.2 $\cos 3\theta$:

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

5.3.3 $\tan 3\theta$:

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

5.4 Formulas for 4θ

$$\sin 4\theta = 4 \sin \theta \cos \theta \cos 2\theta$$

$$\cos 4\theta = 1 - 8 \sin^2 \theta \cos^2 \theta$$

6 Trigonometric Ratios of Sub-Multiple Angles

6.1 Definition

Sub-Multiple Angle: $\theta/2, \theta/3, \theta/4, \dots (\theta/n)$

6.2 Formulas for $\theta/2$ (Half-Angle)

$$\sin \left(\frac{\theta}{2} \right) = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\cos \left(\frac{\theta}{2} \right) = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\tan \left(\frac{\theta}{2} \right) = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

6.3 Alternate Half-Angle Formulas

$$\sin \left(\frac{\theta}{2} \right) = \frac{1 - \cos \theta}{\sin \theta}$$

$$\cos \left(\frac{\theta}{2} \right) = \frac{1 + \cos \theta}{\sin \theta}$$

$$\tan \left(\frac{\theta}{2} \right) = \frac{1 - \cos \theta}{\sin \theta} = \frac{\sin \theta}{1 + \cos \theta}$$

6.4 Formula for $\theta/3$

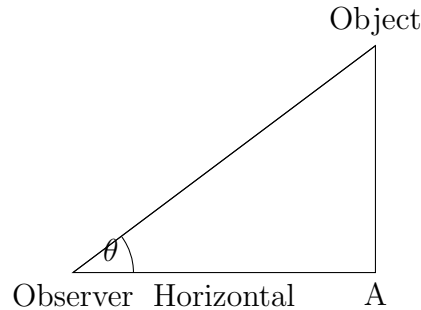
$$\sin\left(\frac{\theta}{3}\right) = 3 \sin\left(\frac{\theta}{3}\right) - 4 \sin^3\left(\frac{\theta}{3}\right)$$

$$\cos\left(\frac{\theta}{3}\right) = 4 \cos^3\left(\frac{\theta}{3}\right) - 3 \cos\left(\frac{\theta}{3}\right)$$

7 Heights and Distances

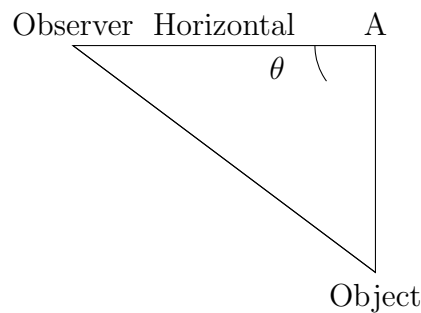
7.1 Angle of Elevation

The angle formed by the line of sight with the horizontal when the object is **above** the observer.



7.2 Angle of Depression

The angle formed by the line of sight with the horizontal when the object is **below** the observer.



7.3 Key Formulas

$$\text{Height} = \text{Distance} \times \tan \theta$$

$$\text{Distance} = \frac{\text{Height}}{\tan \theta}$$

7.4 For Two Objects (Same Side)

$$h = d \times (\tan \alpha - \tan \beta)$$

$$d = \frac{h}{\tan \alpha - \tan \beta}$$

7.5 For Two Objects (Opposite Sides)

$$h = d_1 \tan \alpha = d_2 \tan \beta$$

$$d_1 + d_2 = d$$

7.6 For Heights and Distances from a Point

If angle of elevation from point A is α , from B is β , and $AB = d$:

$$h = d \times \frac{\tan \alpha \tan \beta}{\tan \beta - \tan \alpha}$$

8 Practice Questions (Topic-wise)

8.1 Topic 1: Basic Trig Functions

Q1. In a right triangle ABC, right-angled at B, $AB = 3$ cm, $BC = 4$ cm. Find $\sin A$, $\cos A$, and $\tan A$.

Solution:

$$AC^2 = AB^2 + BC^2 = 3^2 + 4^2 = 9 + 16 = 25$$

$$AC = 5 \text{ cm}$$

$$\sin A = \frac{BC}{AC} = \frac{4}{5}$$

$$\cos A = \frac{AB}{AC} = \frac{3}{5}$$

$$\tan A = \frac{BC}{AB} = \frac{4}{3}$$

Q2. If $\sin \theta = 3/5$, find $\cos \theta$ and $\tan \theta$.

Solution:

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{3}{5}\right)^2 + \cos^2 \theta = 1$$

$$\frac{9}{25} + \cos^2 \theta = 1$$

$$\cos^2 \theta = \frac{16}{25} \rightarrow \cos \theta = \frac{4}{5}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{3/5}{4/5} = \frac{3}{4}$$

Q3. If $\tan \theta = 5/12$, find all other trigonometric ratios.

Solution:

$$\tan \theta = 5/12 = \text{Opposite/Adjacent}$$

$$\text{Let Opposite} = 5k, \text{ Adjacent} = 12k$$

$$\text{Hypotenuse} = \sqrt{25k^2 + 144k^2} = \sqrt{169k^2} = 13k$$

$$\sin \theta = 5/13, \quad \cos \theta = 12/13, \quad \cot \theta = 12/5$$

$$\theta = 13/5, \quad \sec \theta = 13/12$$

Q4. Find the value of $\sin 60^\circ \times \cos 30^\circ + \cos 60^\circ \times \sin 30^\circ$.

Solution:

$$\begin{aligned} &= \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{3}}{2} \right) + \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) \\ &= \frac{3}{4} + \frac{1}{4} = 1 \end{aligned}$$

Q5. Prove that: $\sin^2 \theta + \cos^2 \theta = 1$.

Solution:

In right triangle ABC, $\angle B = 90^\circ$

$$\begin{aligned} \sin \theta &= \frac{BC}{AC}, \quad \cos \theta = \frac{AB}{AC} \\ \sin^2 \theta + \cos^2 \theta &= \frac{BC^2}{AC^2} + \frac{AB^2}{AC^2} \\ &= \frac{BC^2 + AB^2}{AC^2} = \frac{AC^2}{AC^2} = 1 \end{aligned}$$

8.2 Topic 2: Negative Angles

Q6. Find the value of $\sin(-30^\circ) + \cos(-60^\circ)$.

Solution:

$$\begin{aligned}\sin(-30^\circ) &= -\sin 30^\circ = -\frac{1}{2} \\ \cos(-60^\circ) &= \cos 60^\circ = \frac{1}{2} \\ \text{Sum} &= -\frac{1}{2} + \frac{1}{2} = 0\end{aligned}$$

Q7. Evaluate: $\sec(-60^\circ) + \operatorname{cosec}(-30^\circ)$.

Solution:

$$\begin{aligned}\sec(-60^\circ) &= \sec 60^\circ = 2 \\ (-30^\circ) &= -30^\circ = -2 \\ \text{Sum} &= 2 - 2 = 0\end{aligned}$$

Q8. Prove that: $\sin(-\theta) \cos \theta + \cos(-\theta) \sin \theta = 0$.

Solution:

$$\begin{aligned}\text{LHS} &= \sin(-\theta) \cos \theta + \cos(-\theta) \sin \theta \\ &= (-\sin \theta) \cos \theta + \cos \theta \sin \theta \\ &= -\sin \theta \cos \theta + \sin \theta \cos \theta = 0 = \text{RHS}\end{aligned}$$

8.3 Topic 3: Associated Angles

Q9. Simplify: $\sin(90^\circ - \theta) \cos \theta + \cos(90^\circ - \theta) \sin \theta$.

Solution:

$$\begin{aligned}\sin(90^\circ - \theta) &= \cos \theta \\ \cos(90^\circ - \theta) &= \sin \theta \\ &= \cos \theta \cdot \cos \theta + \sin \theta \cdot \sin \theta \\ &= \cos^2 \theta + \sin^2 \theta = 1\end{aligned}$$

Q10. Evaluate: $\sin 150^\circ + \cos 150^\circ$.

Solution:

$$\begin{aligned}\sin 150^\circ &= \sin(180^\circ - 30^\circ) = \sin 30^\circ = \frac{1}{2} \\ \cos 150^\circ &= \cos(180^\circ - 30^\circ) = -\cos 30^\circ = -\frac{\sqrt{3}}{2} \\ \text{Sum} &= \frac{1}{2} - \frac{\sqrt{3}}{2} = \frac{1 - \sqrt{3}}{2}\end{aligned}$$

Q11. Simplify: $\cos(180^\circ + \theta) + \cos(180^\circ - \theta)$.

Solution:

$$\begin{aligned}\cos(180^\circ + \theta) &= -\cos \theta \\ \cos(180^\circ - \theta) &= -\cos \theta \\ \text{Sum} &= -\cos \theta - \cos \theta = -2 \cos \theta\end{aligned}$$

8.4 Topic 4: Compound Angles

Q12. Find the value of $\sin 75^\circ$.

Solution:

$$\begin{aligned}\sin 75^\circ &= \sin(45^\circ + 30^\circ) \\ &= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ \\ &= \left(\frac{1}{\sqrt{2}}\right) \left(\frac{\sqrt{3}}{2}\right) + \left(\frac{1}{\sqrt{2}}\right) \left(\frac{1}{2}\right) \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}}\end{aligned}$$

Q13. Find the value of $\cos 15^\circ$.

Solution:

$$\begin{aligned}\cos 15^\circ &= \cos(45^\circ - 30^\circ) \\ &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\ &= \left(\frac{1}{\sqrt{2}}\right) \left(\frac{\sqrt{3}}{2}\right) + \left(\frac{1}{\sqrt{2}}\right) \left(\frac{1}{2}\right) \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}}\end{aligned}$$

Q14. If $\sin A = 3/5$ and $\cos B = 5/13$, find $\sin(A+B)$.

Solution:

$$\begin{aligned}\sin A &= 3/5 \rightarrow \cos A = 4/5 \\ \cos B &= 5/13 \rightarrow \sin B = 12/13 \\ \sin(A+B) &= \sin A \cos B + \cos A \sin B \\ &= \left(\frac{3}{5}\right) \left(\frac{5}{13}\right) + \left(\frac{4}{5}\right) \left(\frac{12}{13}\right) \\ &= \frac{15}{65} + \frac{48}{65} = \frac{63}{65}\end{aligned}$$

Q15. Prove that: $\sin(A+B) \sin(A-B) = \sin^2 A - \sin^2 B$.

Solution:

$$\begin{aligned}\text{LHS} &= (\sin A \cos B + \cos A \sin B)(\sin A \cos B - \cos A \sin B) \\ &= \sin^2 A \cos^2 B - \cos^2 A \sin^2 B \\ &= \sin^2 A(1 - \sin^2 B) - (1 - \sin^2 A) \sin^2 B \\ &= \sin^2 A - \sin^2 A \sin^2 B - \sin^2 B + \sin^2 A \sin^2 B \\ &= \sin^2 A - \sin^2 B = \text{RHS}\end{aligned}$$

8.5 Topic 5: Multiple Angles

Q16. If $\sin \theta = 3/5$, find $\sin 2\theta$.

Solution:

$$\begin{aligned}\cos \theta &= 4/5 \\ \sin 2\theta &= 2 \sin \theta \cos \theta \\ &= 2 \left(\frac{3}{5} \right) \left(\frac{4}{5} \right) = \frac{24}{25}\end{aligned}$$

Q17. Find the value of $\cos 2\theta$ if $\sin \theta = 1/2$.

Solution:

$$\begin{aligned}\cos 2\theta &= 1 - 2 \sin^2 \theta \\ &= 1 - 2 \left(\frac{1}{2} \right)^2 \\ &= 1 - \frac{1}{2} = \frac{1}{2}\end{aligned}$$

Q18. If $\tan \theta = 3/4$, find $\tan 2\theta$.

Solution:

$$\begin{aligned}\tan 2\theta &= \frac{2 \tan \theta}{1 - \tan^2 \theta} \\ &= \frac{2(3/4)}{1 - 9/16} \\ &= \frac{3/2}{7/16} \\ &= \frac{3}{2} \times \frac{16}{7} = \frac{24}{7}\end{aligned}$$

Q19. Find the value of $\sin 3\theta$ if $\sin \theta = 1/2$.

Solution:

$$\begin{aligned}\sin 3\theta &= 3 \sin \theta - 4 \sin^3 \theta \\ &= 3 \left(\frac{1}{2} \right) - 4 \left(\frac{1}{8} \right) \\ &= \frac{3}{2} - \frac{1}{2} = 1\end{aligned}$$

8.6 Topic 6: Sub-Multiple Angles

Q20. Find $\sin 15^\circ$ using half-angle formula.

Solution:

$$\begin{aligned}\sin 15^\circ &= \sin(30^\circ/2) \\ &= \sqrt{\frac{1 - \cos 30^\circ}{2}} \\ &= \sqrt{\frac{1 - \sqrt{3}/2}{2}} \\ &= \sqrt{\frac{2 - \sqrt{3}}{4}} \\ &= \frac{\sqrt{2 - \sqrt{3}}}{2} = \frac{\sqrt{3} - 1}{2\sqrt{2}}\end{aligned}$$

Q21. Find $\tan 22.5^\circ$.

Solution:

$$\begin{aligned}\tan 22.5^\circ &= \tan(45^\circ/2) \\ &= \sqrt{\frac{1 - \cos 45^\circ}{1 + \cos 45^\circ}} \\ &= \sqrt{\frac{1 - 1/\sqrt{2}}{1 + 1/\sqrt{2}}} \\ &= \sqrt{\frac{\sqrt{2} - 1}{\sqrt{2} + 1}} \\ &= \sqrt{2} - 1\end{aligned}$$

Q22. If $\cos \theta = 1/3$, find $\sin(\theta/2)$.

Solution:

$$\begin{aligned}\sin(\theta/2) &= \sqrt{\frac{1 - \cos \theta}{2}} \\ &= \sqrt{\frac{1 - 1/3}{2}} \\ &= \sqrt{\frac{2/3}{2}} \\ &= \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}\end{aligned}$$

8.7 Topic 7: Heights and Distances

Q23. A ladder 10 m long reaches a window 8 m above the ground. Find the angle of elevation of the ladder.

Solution:

$$\sin \theta = \frac{8}{10} = \frac{4}{5}$$
$$\theta = \sin^{-1} \left(\frac{4}{5} \right) = 53.13^\circ$$

Q24. From a point 100 m away from the foot of a tower, the angle of elevation of the top is 30° . Find the height of the tower.

Solution:

$$\tan 30^\circ = \frac{\text{height}}{\text{distance}}$$
$$\frac{1}{\sqrt{3}} = \frac{h}{100}$$
$$h = \frac{100}{\sqrt{3}} = 57.74 \text{ m}$$

Q25. A man is standing on the top of a building 50 m high. He observes a car at an angle of depression of 30° . How far is the car from the building?

Solution:

$$\tan 30^\circ = \frac{50}{d}$$
$$\frac{1}{\sqrt{3}} = \frac{50}{d}$$
$$d = 50\sqrt{3} = 86.6 \text{ m}$$

Q26. From the top of a lighthouse 150 m high, the angle of depression of a ship is 30° . Find the distance of the ship from the lighthouse.

Solution:

$$\tan 30^\circ = \frac{150}{d}$$
$$d = 150\sqrt{3} = 259.8 \text{ m}$$

Q27. The shadow of a vertical pole is $\sqrt{3}$ times its height. Find the elevation of the sun.

Solution:

$$\text{Shadow} = \sqrt{3} \times \text{height}$$

$$\tan \theta = \frac{\text{height}}{\text{shadow}} = \frac{1}{\sqrt{3}}$$

$$\theta = 30^\circ$$

Q28. From a point on the ground, the angles of elevation of the top of a tower and a flagstaff on top of the tower are 30° and 45° respectively. If the height of the tower is 100 m, find the height of the flagstaff.

Solution:

$$\text{Let height of flagstaff} = x$$

$$\tan 30^\circ = \frac{100}{d} \rightarrow d = 100\sqrt{3}$$

$$\tan 45^\circ = \frac{100 + x}{d}$$

$$1 = \frac{100 + x}{100\sqrt{3}}$$

$$100\sqrt{3} = 100 + x$$

$$x = 100(\sqrt{3} - 1) = 73.2 \text{ m}$$

9 HPU Exam Important Questions (Unit-III Only)

Based on HPU BCA-0101 Previous Year Papers

9.1 Section-A (5 Marks Each)

- Q1.** If $\sin \theta = 3/5$, find all other trigonometric ratios.
Q2. Find the value of $\sin 75^\circ$.
Q3. Prove that: $\sin 2\theta = 2 \sin \theta \cos \theta$.
Q4. From a point on the ground, the angle of elevation of the top of a tower is 30° . If the tower is 100 m high, find the distance of the point from the tower.
Q5. Evaluate: $\sin 120^\circ + \cos 210^\circ$.
Q6. If $\tan \theta = 3/4$, find the value of $\sin 2\theta$.
Q7. Find the value of $\cos 15^\circ$.
Q8. Prove that: $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$.

9.2 Section-B (5 Marks Each)

- Q9.** If $\sin A = 4/5$ and $\cos B = 5/13$, find the value of $\sin(A+B)$.
Q10. Find the value of $\tan 22.5^\circ$.
Q11. Prove that: $\sin(90^\circ - \theta) = \cos \theta$.
Q12. A man standing on the top of a cliff 100 m high observes a boat at an angle of depression of 30° . Find the distance of the boat from the foot of the cliff.
Q13. If $A + B = 60^\circ$, prove that $\sin(A+B) = \sin A \cos B + \cos A \sin B$.
Q14. Find the value of $\tan 75^\circ$.
Q15. If $\tan \theta = 5/12$, find the value of $\sin 2\theta$.
Q16. Prove that: $(1 + \tan \theta)^2 = \sec^2 \theta + 2 \tan \theta$.

9.3 Section-C (10 Marks Each)

- Q17.** Prove that: $\sin(A+B) \sin(A-B) = \sin^2 A - \sin^2 B$.
Q18. A man observes the angle of elevation of the top of a tower to be 30° . He walks 100 m towards the tower and finds the angle of elevation to be 60° . Find the height of the tower.
Q19. Prove that: $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$.
Q20. From the top of a building 50 m high, the angle of depression of the top and bottom of a tower are 30° and 60° respectively. Find the height of the tower.
Q21. Prove that: $\tan 3\theta = (3 \tan \theta - \tan^3 \theta)/(1 - 3 \tan^2 \theta)$.

9.4 Section-E (Compulsory - 1 Mark Each)

- Q22.** Write the value of $\sin 0^\circ$.
Q23. What is the value of $\tan 45^\circ$?
Q24. What is the value of $\cos 30^\circ$?
Q25. Write the formula for $\sin(A+B)$.
Q26. What is the formula for $\sin 2\theta$?
Q27. What is the maximum value of $\sin \theta$?
Q28. What is the minimum value of $\cos \theta$?
Q29. What is the value of $\sin(-30^\circ)$?
Q30. Write the formula for $\cos 2\theta$.
Q31. What is the value of $\sec 60^\circ$?

Q32. What is the reciprocal of $\sin \theta$?

Q33. What is $\sin 90^\circ$?

10 Quick Revision Formulas

10.1 Fundamental Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

10.2 Negative Angles

$$\sin(-\theta) = -\sin \theta, \quad \cos(-\theta) = \cos \theta, \quad \tan(-\theta) = -\tan \theta$$

10.3 Compound Angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

10.4 Multiple Angles

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

10.5 Half-Angle Formulas

$$\sin \left(\frac{\theta}{2} \right) = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\cos \left(\frac{\theta}{2} \right) = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\tan \left(\frac{\theta}{2} \right) = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

θ	0°	30°	45°	60°	90°
sin	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
cos	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞

10.6 Standard Values

10.7 Heights and Distances

$$\tan \theta = \frac{\text{height}}{\text{distance}}$$

$$\text{height} = \text{distance} \times \tan \theta$$

$$\text{distance} = \text{height} \times \cot \theta$$

Exam Tips for Unit-III

1. **Memorize Standard Values** - 0° , 30° , 45° , 60° , 90°
2. **ASTC Rule** - Remember which ratios are positive in each quadrant
3. **Compound Angles** - $\sin(A \pm B)$, $\cos(A \pm B)$ formulas - Very important
4. **Multiple Angles** - $\sin 2\theta$, $\cos 2\theta$, $\sin 3\theta$ formulas
5. **Heights & Distances** - Draw diagrams and use tan formula

— End of Unit-III Notes —